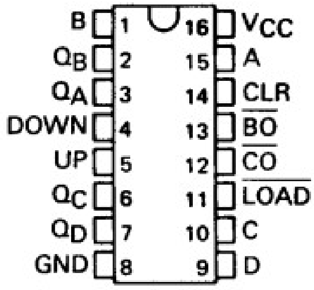
4-BIT SYNCHRONOUS UP/DOWN BINARY COUNTERS

DECEMBER 1982 – REVISED OCTOBER 2003

DESCRIPTION

The 74193 is a universal, synchronous 4-bit binary counter that can count both forward (up) and backward (down). Since the counter operates synchronously, all flip-flops switch simultaneously with the rising edge of the clock signal. This prevents switching spikes (glitch-free operation) which can occur in asynchronous counters (such as the 7490). Unlike other counters, the 74193 has separate clock inputs for counting up and counting down, as well as a direct clear input (clear).



Pinout Diagram

Name	Function
A,B,C,D	Data input bits 1 to 4 (for start value)
Q_A, Q_B, Q_C, Q_D	Data output bits 1 to 4
CLEAR	Asynchronous reset (High = sets all outputs to Low)
UP	Clock input counting up (rising edge)
DOWN	Clock input counting down (rising edge)
~LOAD	Load data (Low = load inputs A-D)
~CARRY (~CO)	Overflow output (gives low-impulse when overflowing from 15 to 0)
~BORROW (~BO)	Underflow output (gives low-impulse when underflowing from 0 to 15)

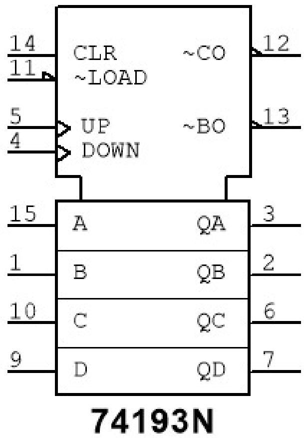
FUNCTIONALITY

Counting (Up / Down): The counter responds to a transition from low to high (rising edge) at the clock inputs. To count upwards, the clock signal must be applied to the UP-input while the DOWN-input must remain at high.
To count backward, the clock signal must be applied to the DOWN-input while the UP-input must remain at high.

Loading a start value (Preset):
A start value can be adopted through a low signal at the ~LOAD input. To do this, the values present at the data inputs (A, B, C, D) are transferred directly to the outputs (QA, QB, QC, QD), completely independent of the clock signal.

Reset (Clear):
A High-signal at the CLEAR-input immediately forces all outputs to a logical 0. This occurs asynchronously, meaning the counter does not wait for the next clock pulse. The CLEAR-input has the highest priority.

Cascading (Extension):
The ~CARRY and ~BORROW outputs are used to connect several 74193 modules in a row (e.g., to count up to 255). ~CARRY generates an impulse during upward counting when the maximum value (15) is exceeded. ~BORROW generates an impulse during downward counting when the counter goes below zero.



Logic Symbol

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